

Modeling a Sudoku Puzzle

$P_{i,j,n}$: the number at the cell (i,j) is n .

$\varphi := \bigwedge_k P_{i_k j_k n_k}$ (known). $\varphi' = \bigwedge_{i,j} \bigwedge_{n \neq n'} \neg (P_{i,j,n} \wedge P_{i,j,n'})$. (Certainty)

$\phi_1 := \bigwedge_n (\bigwedge_i (\bigvee_j P_{i,j,n}))$ (for each and each row contains at least one or).

$$\phi_1 \wedge \varphi'$$

$\phi_2 := \bigwedge_n (\bigwedge_j (\bigvee_i P_{i,j,n}))$ (for each n , each column contains at least one).

$$\phi_2 \wedge \varphi'$$

Define $A_1 \sim A_9 : k: 1 \sim 9$.

$\phi_3 = \bigwedge_n (\bigwedge_k (\bigvee_{(i,j) \in A_k} P_{i,j,n}))$ (for each n , every 3×3 blocks contains at least one).

$$\phi_3 \wedge \varphi'$$

$$\varphi \wedge \varphi' \wedge \phi_1 \wedge \phi_2 \wedge \phi_3.$$